KENYA MEDICAL TRAINING COLLEGE
FACULTY OF REHABILITATIVE SCIENCES
DEPARTMENT OF MEDICAL ENGINEERING

FINAL QUALIFYING EXAMINATION
FOR
CERTIFICATE IN MEDICAL ENGINEERING TECHNOLOGY

PAPER: MATHEMATICS

DATE: 13th June 2022 **TIME: 3 HOURS (9:00AM – 12:00 PM)**

INSTRUCTIONS

1. This paper consists of:
   - Section 1 (10 Short Answer Questions)
   - Section 2 (3 Long Answer Questions)
2. Attempt **ALL** Questions
3. Write the **EXAMINATION NUMBER** given on all the answer sheets provided and on the question paper.
4. Ensure that all examination answer scripts are handed in at the end of the examination
5. Ensure you sign the examination register provided

EXAMINATION NUMBER .....

SECTION 1: SHORT ANSWER QUESTIONS (40 MARKS)

1. The sum of eight terms of an arithmetic progression is 42 and the common difference is 2. Determine the first term of the series. (4 marks)
2. The 1st , 80th and 120th numbers of an arithmetic progression are $2\frac{1}{2}$,121 and 181 respectively. Determine the sum of all the terms. (4 marks)
3. The first term of a geometric series is 10, and the fourth term is 33.75. Determine the sixth term and the tenth term, each correct to 5 sgnificant figures. (4 marks)
4. From the first principles, differentiate $y = 4x$ (4 marks)
5. Differentiate $\sin(6x + 1)$ (4 marks)
6. Find the value of the mean, median and mode of the following set of data; $52, 51, 52, 59, 60, 58, 54, 53$ and 50 (4 marks)
7. If matrix $A = \begin{pmatrix} 2 & 4 \\ 1 & -3 \end{pmatrix}$, $B = \begin{pmatrix} 3 & -7 \\ 4 & -5 \end{pmatrix}$. Determine $A.B$ (4 marks)
8. Solve $\int_1^2 (x^2 - 3)dx$ (4 marks)
9. Express 492.731_{10} in octal form. (4 marks)
10. Express as a single fraction $\frac{4}{x+1} + \frac{x}{1+x} - 3$ (4 marks)

SECTION 2: LONG ANSWER QUESTIONS (60 MARKS)

11. a) Determine the value of arbitrary constant of integration if $m = \int \frac{t^2-4t}{t}\, dt$ when $t = 2$ and $m = 3$. (5 marks)
    b) Prove that $\frac{1+tan^2 B}{1+cot^2 B} = tan^2 B$ (5 marks)
    c) The total surface area of a closed cylinder is $125cm^2$. Calculate the radius of the cylinder if its height is $5.24cm$. (10 marks)

12. a) Solve the equation $3^{x+1} = 2^{2x-3}$ for the value of x. (5 marks)
    b) Find the differential coefficient of $tanx$. (10 marks)
    c) If $sin\theta = \frac{5}{13}$, determine;
       i. $cos\theta$
       ii. $tan\theta$
       iii. $sec\theta$ (5 marks)

13. a) In a triangle ABC, angle $A = 53^0, B = 61^0$ and the length of $a = 12.60cm$. Find the unknown sides and angle. (10 marks)
    b) Using Binomial expansion theorem, determine the value of $(1.05)^3\ (0.98)^4$ correct to 5 significant figures. (10 marks)